

THREAD CREATION USING PTHREAD

C PROGRAM:

```
#include <stdio.h>

#include <stdlib.h>

#include <pthread.h>

#include <unistd.h>

void *thread_function(void *arg){

    int i;

    for(i = 1; i <= 5; i++)

    {

        printf("Thread Executing : %d\n", i);

        sleep(1);

    }

    pthread_exit(NULL);

}

int main()

{

    pthread_t t1, t2;

    pthread_create(&t1, NULL, thread_function, NULL);

    pthread_create(&t2, NULL, thread_function, NULL);

    pthread_join(t1, NULL);

    pthread_join(t2, NULL);

    printf("All Threads Completed\n");

    return 0;

}
```

SHELL SCRIPTING:

```
#!/bin/bash
```

```
task1()
```

```
{
```

```
for i in 1 2 3 4 5
```

```
do
```

```
echo "Thread 1 : $i"
```

```
sleep 1
```

```
done
```

```
}
```

```
task2()
```

```
{
```

```
for i in 1 2 3 4 5
```

```
do
```

```
echo "Thread 2 : $i"
```

```
sleep 1
```

```
done
```

```
}
```

```
task1 &
```

```
task2 &
```

```
wait
```

```
echo "All Threads Completed"
```

OUTPUT:

```
(kali㉿Anbuchelvan)-[~]  
$ nano threading.sh
```

```
(kali㉿Anbuchelvan)-[~]  
$ chmod +x threading.sh
```

```
(kali㉿Anbuchelvan)-[~]  
$ bash threading.sh
```

Thread 1 : 1

Thread 2 : 1

Thread 1 : 2

Thread 2 : 2

Thread 1 : 3

Thread 2 : 3

Thread 1 : 4

Thread 2 : 4

Thread 1 : 5

Thread 2 : 5

All Threads Completed